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INDUSTRIAL PIPELINE OPERATIONS MANUAL: HOOP STRESS EVALUATION

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1. APPLICABILITY

This standard applies strictly to all high-pressure gas and liquid transmission lines composed of API 5L Grade X-52 carbon steel.

2. MATERIAL SPECIFICATIONS (API 5L GRADE X-52)

- Specified Minimum Yield Strength (SMYS): 358 MPa
- Ultimate Tensile Strength: 455 MPa

3. PRESSURE & STRESS COMPUTATION (BARLOW'S FORMULA)

Hoop stress generated by internal fluid pressure is calculated using Barlow's Equation:

- Formula: $\text{Hoop_Stress_MPa} = (\text{Pressure_MPa} * \text{Internal_Diameter_mm}) / (2 * \text{Wall_Thickness_mm})$
- Unit Conversion: 1 Bar = 0.1 N/mm² (MPa). Pressure supplied in Bar must be multiplied by 0.1.

4. SAFETY FACTOR & THRESHOLD PROTOCOL

In compliance with pipeline safety regulations, operating hoop stress must never exceed 72% of the SMYS:

- Allowable Operating Hoop Stress = $0.72 * 358 \text{ MPa} = 257.76 \text{ MPa}$.

5. EMERGENCY ACTION LOGIC

- IF Calculated_Hoop_Stress $\leq 257.76 \text{ MPa}$: System is COMPLIANT.
- IF Calculated_Hoop_Stress $> 257.76 \text{ MPa}$: NON-COMPLIANT. Trigger emergency depressurization (SOP-12-B).